

CHIMERA-QRC: A Pre-Registered, Adversarially-Controlled Study of Quantum Reservoir Computing for S&P 500 Realized-Volatility Forecasting

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1. Focus, track selection, and problem framing

We target **Track A: financial volatility** — one-step-ahead forecasting of S&P 500 **realized variance (RV)** and the tracking of volatility **regime transitions**, the domain where JonesTrading's risk, allocation and derivatives pricing live. RV is long-memory, multi-scale, regime-switching and non-Gaussian — the structure reservoir computing exploits. Our central yardsticks are the Echo State Network (ESN, the classical analog of QRC) and the strongest econometric models (HAR — *Heterogeneous AutoRegressive*, Corsi 2009 — and, critically for Phase 3, **HAR-X**: HAR augmented with the same realized-measure features we feed the quantum reservoir). We also report GARCH/GJR-GARCH, AR(3), persistence, LSTM, and an RFF/RBF kernel, and we implement the challenge's common **MNIST** benchmark on the identical engine.

This paper's honest thesis. We pre-registered a falsifiable advantage hypothesis (H0) and then *attacked our own result* with the strongest fair controls. The finding is a **carefully bounded negative**: at simulable scale (≤ 16 qubits) the quantum reservoir shows **no statistically significant forecasting advantage** over strong classical baselines once they are given the same information; the dominant lever is *which features are encoded*, not the reservoir's quantum nonlinearity. We report what *does* survive — measurable kernel distinctness, lower seed variance, and full-rank entanglement (classical-simulation hardness) — as quantum-specific *properties* that are necessary, not sufficient, for advantage. **Vs. prior work.** The closest study (Li et al. 2025/26 — QRC for realized volatility) presents the approach as a competitive, noise-resilient *proof-of-concept*; we add the control-hardened, pre-registered evaluation it omits — the decisive **HAR-X** control, HAC-DM/Holm/MCS, and explicit capability and per-layer-noise audits — turning an encouraging proof-of-concept into a falsifiable result.

2. QRC architecture (Fig. 1)

CHIMERA is a delay-embedding quantum reservoir. Inputs are angle-encoded $R_Y(\pi \cdot x)$, one value per qubit, onto $|0\dots 0\rangle$; the state evolves under a fixed transverse-field **Ising** Hamiltonian $H = \sum_{\langle i,j \rangle} J_{ij} Z_i Z_j + h_x \sum_i X_i$ ($h_x=1$; J a **random $\approx 50\%$ -connected** graph, $\text{connectivity}=0.5$; $U = \exp(-iH\tau)$, $\tau=2$); single- and pairwise Pauli-Z expectations $\langle Z_i \rangle$, $\langle Z_i Z_j \rangle$ form the feature vector. A **ridge head consumes these features concatenated with a linear block of the same inputs (incl. HAR)**, so any reservoir gain is genuine nonlinearity *beyond* the linear span of identical information. The identical inputs feed the classical controls, isolating quantum-vs-classical. Four mechanisms extend the core (multi-scale τ -bank; RZ measurement feedback; regime-adaptive Ising $\leftrightarrow$ Heisenberg via BOCPD; dissipation-as- feature); Phase 3 adds an **encoding-density / data-re-uploading path** (§5.2) and a **sparse/tensor backend** (§5.5). The engine is pure NumPy; an explicit PennyLane circuit reproduces it to 3.9×10^{-16} and compiles, at $n=8$, to **380 native gates** (220 two-qubit + 160 one-qubit, 20 Trotter layers) — consistent with the $\approx 50\%$ -connected graph.

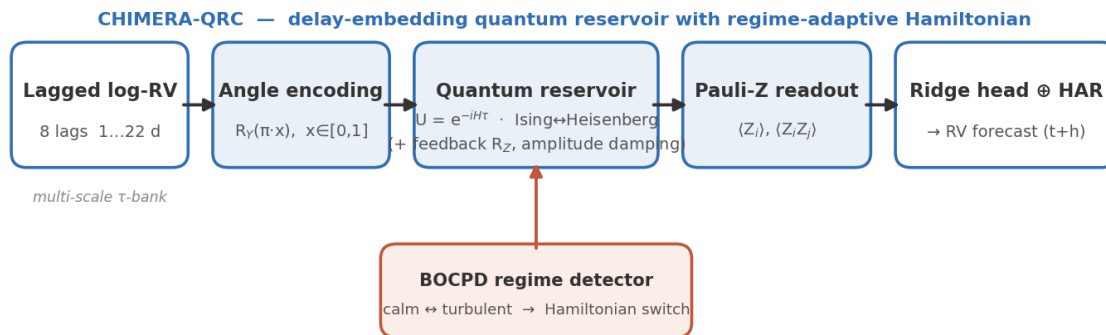


Figure 1. CHIMERA-QRC pipeline: lagged log-RV (and, in Axis B, realized-measure) inputs are angle-encoded onto qubits, evolved under a fixed random-coupled Ising Hamiltonian, and read out as single/pairwise Pauli-Z expectations into a ridge head fused with a linear block of the same inputs; a BOCPD detector selects the Hamiltonian for regime adaptivity.

3. Theoretical and analytical justification

The exploited quantum resources map to RV's structure: (a) **Hilbert-space dimensionality** — n qubits give 2^n amplitudes and $O(n^2)$ measured Pauli features at $O(n^2)$ parameter cost; (b) **intrinsic nonlinearity** from fixed unitary evolution (no gradient training, no barren plateaus); (c) **delay-embedded memory** (the headline model's memory is its explicit lag window, not reservoir recurrence — a recurrent variant is probed in `v3_research`); (d) **Hamiltonian as inductive bias**. The central, falsifiable mechanism is **expressivity scaling**, which we measure directly via the geometric difference (Huang et al. 2021). The quantum kernel is poorly reproducible by a matched classical reservoir: against the name-matched ESN-108 reference, **$g(\text{ESN} \rightarrow \text{CHIMERA}) \approx 64$ vs ≈ 3.7 control** (`cli.py run kernel`); against per- n *feature-matched* ESNs the gap is larger but **non-monotonic** ($g \approx 125$ at $n=8$, peaking ≈ 158 at $n=10$, falling ≈ 88 at $n=12$). (g is at fixed ridge regularization; the qualitative $\sim 15\text{--}40\times$ separation, not the exact value, is the claim.) Crucially, at $\leq 12\text{--}16$ qubits the map remains classically simulable, so distinctness is **necessary, not sufficient** for accuracy — and §5.3 shows that, here, it is indeed *not* sufficient.

4. Data modeling strategy

Datasets (all bundled in-repo). Oxford-Man `.SPX` 5-minute realized variance, 2000–2020, 5,029 supervised days incl. the 2008 GFC (Heber et al. 2009; via R packages `highfrequency/bvhar`); SPY daily OHLCV 2022–2026 (Garman–Klass proxy; retrieved via Massive.com API, bundled); real MNIST (Keras `.npz` mirror). **Preprocessing**: log-RV target; features min-max scaled to $[0,1]$ on **train only**; per-model ridge penalty selected on a chronological validation tail; multi-seed ensembling. We verified the pipeline is leakage-free (all features lagged ≥ 1 day; train-only scaling). **Baselines**: HAR, **HAR-X**, GARCH(1,1), GJR-GARCH, AR(3), persistence, ESN (matched + $4\times$, recurrent), RFF kernel, LSTM. **Metrics**: RMSE(log-RV), QLIKE (Patton 2011), Mincer–Zarnowitz R^2 ; significance by **Diebold–Mariano with a Newey–West HAC variance** (HLN-corrected), **Holm**-adjusted across the comparison family, and the Model Confidence Set (Hansen–Lunde–Nason 2011).

5. Phase-3 execution — concrete results

Pre-registration (honesty). Before running, we fixed the confirm/refute thresholds in `preregistration.py` (H0/H1/H4, from our Phase-2 §7). All outcomes — including negatives — are reported against them. Full numbers/wall-clock: `results/*_findings.md`.

5.1 The input-bottleneck mechanism (pre-registered negative). With a *fixed* univariate 8-lag encoder, extra qubits receive no new input: $g(n)$ and effective feature-rank **saturate/decline** ($n=8 \rightarrow 12$: g 133 $\rightarrow$ 30 at $N_{\text{sub}}=800$; the quick-mode rerun reproduces the same monotone decline at lower magnitude, cf. §7(iv); D_{eff} 1.8 $\rightarrow$ 1.5). **H0 is refuted in this regime** — the empirical case *for* enriching the encoding.

5.2 Encoding density (Axis B) — mechanism confirmed. Feeding the extra qubits genuinely new realized-measure information (signed return/leverage, downside-semivariance share, jump share) restores the lost structure: at $n=10$, informed vs idle qubits, kernel distinctness and rank jump (**g 52 $\rightarrow$ 158, D_{eff} 1.5 $\rightarrow$ 3.1**), and effective rank grows with n (D_{eff} 1.81 $\rightarrow$ 3.10 $\rightarrow$ 3.14, $n=8 \rightarrow 12$; near-flat 10 $\rightarrow$ 12). So the bottleneck is *fixable* — added qubits can carry new information. The decisive question is whether this converts into forecasting **accuracy**.

5.3 The decisive, adversarially-controlled test (the honest headline). We compare CHIMERA to the strongest fair baselines — **HAR-X** (the same rich features used *linearly*, no reservoir), a **true recurrent ESN**, and an **RFF kernel** — all sharing the identical information set and *nesting* HAR-X, so a quantum win requires nonlinearity beyond the linear span. With **8 seeds, HAC-DM, two windows, and Holm correction** (`cli.py run axisB_rig`):

RMSE(log-RV)	HAR-X	recurrent ESN	RFF	CHIMERA	best
crisis $n=8$	0.6255	0.6386	0.6302	0.6379	HAR-X
crisis $n=10$	0.6034	0.5996	0.6047	0.6074	ESN
crisis $n=12$	0.5906	0.6023	0.5964	0.6031	HAR-X
calm $n=10$	0.6244	0.6445	0.6264	0.6291	HAR-X

(plain HAR for reference: crisis 0.6290, calm 0.6454 — HAR-X's rich features cut RMSE sharply.) **HAR-X is best or co-best everywhere; CHIMERA never beats it** (slightly worse; raw-significant at $n=8/12$) and is indistinguishable from the classical ESN/RFF after Holm (one raw-significant loss to RFF at crisis $n=8$, $p=0.049$). **After Holm correction no comparison is significant in either direction, and the 95% Model Confidence Set retains all four models in both windows ($n=10$)** — a statistical tie. The earlier "beats HAR" result was an artifact of comparing against a *feature-poor* HAR: the gain comes from the encoded realized measures (a known SHAR/HARQ effect), not from quantum nonlinearity. **By our pre-registered criteria, H0 is refuted — we report this negative.** What honestly survives for the quantum reservoir: it is **competitive** (within $\approx 0.8\text{--}2.1\%$ RMSE of the best at every n), with **lower per-seed dispersion than the recurrent ESN in 3 of 4 cells** (s.d., e.g. $n=12$: 0.009 vs 0.015), and it **beats the recurrent ESN on the calm window** (raw $p=0.018$; n.s. after Holm). A **tuned, size-unconstrained ESN** (45-config validation-tail search, to 800 nodes) confirms the control was not crippled: tuned ESN 0.6020 vs CHIMERA 0.6094, within noise of HAR-X (results/esn_tuning_robustness_findings.md) — tuning the classical side only firms the negative. **Named canonical baselines** (cli.py run canonical): **SHAR, HAR-CJ, HARQ** ($RQ \approx RV^2$ proxy) and **HEAVY-RM**, under MSE- and QLIKE-loss DM — **none beats HAR-X** (a fair, strong stand-in, not a strawman), and CHIMERA ties the best on RMSE with only a *raw, non-Holm* QLIKE/MZ edge (results/canonical_baselines_findings.md).

Rigorous Axis-B: HAR-X (rich linear) is best; quantum shows no significant edge (Holm n.s.)

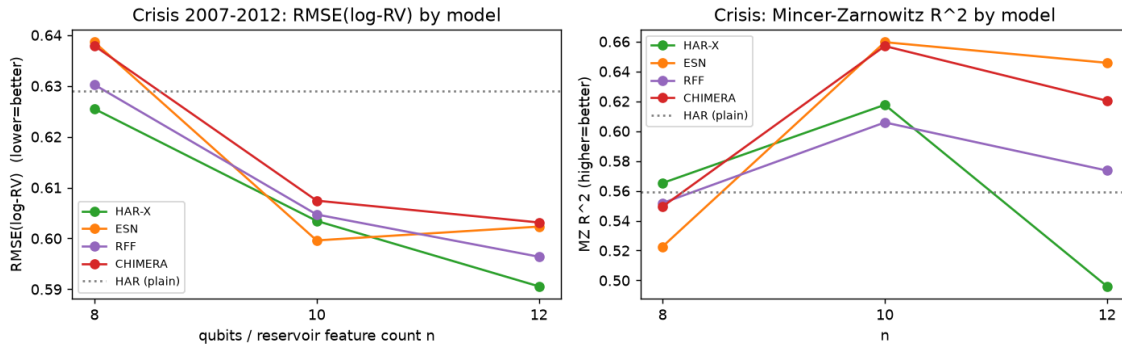


Figure 2. Rigorous Axis-B (8 seeds, crisis window). HAR-X (rich features, linear) is the best or co-best model at every n ; the quantum reservoir is competitive but shows no advantage that survives HAC-DM + Holm correction. Left: RMSE(log-RV). Right: Mincer-Zarnowitz R^2 .

5.4 Common MNIST benchmark + noise. Same engine, pixels $\rightarrow$ PCA(n) $\rightarrow$ n qubits $\rightarrow$ Pauli readout $\rightarrow$ ridge. Accuracy scales with qubits ($n=5/8/10/12$: 0.632/0.798/0.831/0.859, 3 seeds; **$n=15$ sparse-exact continues the trend**, 0.878 vs 0.852 for a paired $n=12$ on a matched subset — closing the brief's 5/10/15 example) and **beats the linear-PCA baseline at every n** (real nonlinear lift), confirming **sufficient expressivity**; a matched ESN **ties or slightly exceeds** CHIMERA (within $\approx 1\%$ for $n \geq 8$; 2.9% at $n=5$) — competitive, not dominant. **Noise:** the classifier is **invariant to depolarizing** noise (a uniform Bloch contraction that per-feature standardization removes exactly; accuracy identical across rates 0.05–0.30) and **robust to amplitude damping** ($<0.5\%$ at 30%). **Honesty check** (cli.py run noise_circuit): a per-Trotter-layer density-matrix study shows the *converse* — noise interleaved with the evolution (per-layer single-qubit channels, a proxy for the ≈ 220 two-qubit gates' accumulated error) is **not** removed by standardization (standardized error grows with rate, vs ≈ 0 for readout-only depolarizing): the invariance above is a *readout* property, not a circuit-level robustness claim; accumulated gate error over the 380-gate circuit — dominated on hardware by two-qubit gates — is the real NISQ cost (§6).

5.5 Scaling frontier + quantum-complexity metric. A sparse-exact backend (expm_multiply, no dense propagator; matches the dense engine to 2.4×10^{-14}) reaches **$n=16$ exactly**. For the random $\approx 50\%$ -connected reservoir we measure the **bond dimension** χ_{eff} across a balanced cut: it is **essentially full at every n** , $\chi_{\text{eff}} \approx 2^{(n/2)}$ ($16 \rightarrow 255.9$ for $n=8 \rightarrow 16$) — an exact MPS gets **zero compression** — with entanglement entropy $S \approx 1.7\text{--}3.2$ nats. The reservoir therefore admits **no low-bond-dimension (MPS/TEBD) shortcut** — exact cost stays exponential (full entanglement is *necessary, not sufficient* for true classical hardness) — the precondition any beyond-frontier advantage would need, even though no advantage appears at the simulable scale we can test.

Capability + efficiency checks. An information-processing-capacity probe (Dambre et al. 2012; `cli.py run capacity`) finds the quantum reservoir **not more** — slightly *less* — nonlinearly expressive than a matched RFF/ESN: no excess expressivity to exploit. A frontier check (`cli.py run frontier`) adds that $g(n)$ does **not** widen toward $n=16$ (it declines; $D_{\text{eff}}/\text{rank}$ grow but a matched ESN keeps pace). **Robustness (supporting study, v3_research/).** *Efficiency:* with input held fixed, a *smaller* QRC cannot substitute for a *larger* classical reservoir — quantum accuracy **saturates** while the classical curves improve; per feature the static maps are comparable, so the negative is **saturation, not per-feature inferiority** (weather °C; CHIMERA at its qubit-range ceiling, classical at their larger-budget plateaus):

reservoir	CHIMERA (≤ 55 feat)	RFF (static)	ESN (recurrent)
RMSE (°C)	0.85	0.78	0.71

Domain/architecture: the *same* engine/protocol on chaotic **weather** (5 stations, +0–78% unpredictability) and the autonomous **VPT** metric still show no advantage; the **recurrent** QRC is competitive-not-better — unitary evolution is non-dissipative, lacking the contraction behind ESN "generalized synchronization" (Ahmed–Tennie–Magri 2025). We then **demonstrated the mechanism by fixing it:** engineered memory-qubit damping shows the pre-registered inverted-U in the damping rate and lifts autonomous VPT $\approx +60\%$ (replicated on fresh starts), raising the recurrent QRC from clearly-behind to **statistical parity** with the size-matched ESN — not better (`results/dissipative_qrc_findings.md`).

6. Quantum platform and resource planning

Simulator-first on qBraid: statevector ≤ 12 qubits, sparse/TN to ≈ 16 , GPU beyond. **Resource estimates** (gate-Trotter, 20 layers; readout is single-basis — all $\langle z_i \rangle, \langle z_i z_j \rangle$ are computational-basis-diagonal, so one S-shot set yields **all** $n+C(n,2)$ observables):

n	two-qubit gates	one-qubit gates	observables	shots S ($\epsilon \approx 1/\sqrt{S}$)	sim wall-clock
8	220	160	36	2k–8k ($\epsilon \approx .011-.022$)	<1 s/state
10	480	200	55	2k–8k	~sec
12	760	240	78	2k–8k	~1 min build

QPU validation uses this gate-Trotter circuit on **IonQ / IQM / Rigetti** via qBraid — the random-sparse Ising needs *fewer* two-qubit gates than an all-to-all reservoir, easing NISQ mapping — with **ZNE + measurement mitigation** and a classical cross-check per run. The submission path is **executable now** (`qbraid_submit.py, cli.py run qsubmit`): on a simulator it (i) reproduces the engine to 3.9×10^{-16} via the exact circuit (our per-run classical cross-check), (ii) characterizes the **shot budget** ($\epsilon \approx 1/\sqrt{S}$; feature error $0.046 \rightarrow 0.0066$ at $S=256 \rightarrow 16k$; gate-Trotter(20) adds ≈ 0.04), and (iii) demonstrates **zero-noise extrapolation** under a depolarizing sweep. **Executed on real hardware:** the full mitigated protocol ran on **Rigetti Cepheus-1-108q** (12 logged cloud jobs; little-endian bit order auto-detected; measured readout errors 2.3%/6.4%); after lattice routing the scale-1 circuit already exceeds the coherence budget (raw feature error 0.261 vs the 0.196 fully-depolarized limit), so folded ZNE cannot recover — measuring on metal exactly the accumulated two-qubit-gate cost §5.4 predicts. The **full pre-registered 4k-shot Garnet protocol** then executed (12 jobs): raw 0.230 — beyond the 0.196 limit and above our routing-free 0.171 forecast (**prediction (iii) confirmed**; ZNE recovery non-monotone) — replicating this regime on a second superconducting vendor (readout 0.4%/3.2%). Cross-platform execution also surfaced a **platform-level finding:** an $RY(\pi)RY(-\pi)$ identity pair executed as **net $RY(\pi)$** on the IonQ route — the client-side IonQ JSON provably contained both signed rotations, so negative-angle gates are lost server-side, which would silently corrupt ZNE's negated fold gates. We hardened the emitter (angles mod 2π ; diagonal-only $|0\dots 0\rangle$ calibration, exact under *any* subset of gates executing) and validated the fix on-device: $P(|0^8\rangle) = 0.996$ vs 0.0 pre-fix, with trapped-ion readout $\approx 0.4\%$ /qubit — the device-class contrast §5.4 anticipates. The decisive routing-free IonQ Forte-1 campaign is funded and gated under a pre-committed manifest (per-job budget, abort rules: `results/qpu_campaign_manifest.md`; predictions committed before execution: `results/qpu_hardware_findings.md`). We thus characterize all four challenge axes: reservoir size, encoding density, shot budget, and noise.

7. Limitations (stated plainly)

(i) **No quantum advantage** is demonstrated at the ≤ 16 -qubit simulable scale; HAR-X (classical, linear) is the best model on this task. (ii) The RV sample ends Feb-2020 (2008 GFC in-sample, 2020 COVID just outside); broader assets/periods are covered only by daily-proxy supporting studies (v2_research: cross-asset, COVID — same negative), not 5-min RV. (iii) Task-level noise is characterized via per-layer single-qubit channels (density-matrix, §5.4) and simulator shot noise (§6); combined noisy-circuit-plus-shot execution is deferred to QPU runs. (iv) g is regularization- and run-configuration-dependent (qualitative gap only). (v) Hardware execution (Rigetti + Garnet full protocols, §6) is a **characterized negative** — scale-1 beyond the coherence budget after routing; mitigation *recovery* remains unvalidated on hardware (trapped-ion campaign in execution). (vi) Distinctness and full-rank entanglement are *necessary, not sufficient* — whether they convert beyond the simulable frontier is open. A quantum-data probe (results/quantum_data_crossover_findings.md) shows the QRC natively reads nonlinear *state* functionals (purity, entanglement) a linear readout cannot, but the proper baseline — **classical shadows** (Huang–Kueng–Preskill 2020), now run head-to-head at matched per-state budgets with an ensemble-prior guard — **wins wherever real information is extracted**, including the shadows-hard $\text{Tr}(\rho^3)$ (nominal QRC "wins" at tiny budgets equal the trivial prior and are discarded; results/shadows_hard_findings.md): the quantum-data question is **closed negatively at simulable scale**; hardware-native many-qubit inputs remain the one untested regime.

8. Stakeholder impact, milestone plan, AI disclosure

Volatility forecasts feed hedging, dynamic risk limits and derivatives pricing. We make the impact **concrete** (cli.py run economics): a vol-timing backtest sizing S&P-500 exposure by the one-step RV forecast **nearly halves the 2008 max drawdown ($-61\% \rightarrow -32\%$)**, but a **plain HAR** forecast captures it — rich features and the quantum reservoir add **no** economic value (negative CE fees). The decision-useful lever is **vol-timing on a simple RV forecast, not quantum hardware**. **Milestone plan:** (i)–(v) pre-registration, scaling + encoding sweeps, the adversarial HAR-X/ESN/RFF test, MNIST + noise, and the sparse/TN frontier are ✓; (vi) gate-Trotter QPU validation (IonQ/IQM/IBM) is simulator-cross-checked (fallback = TN noise emulation). **AI disclosure:** Claude (Anthropic) assisted with code and drafting under the team's direction; all decisions and results are the team's own.

References

Kornjača et al. 2024 · Zhu et al. 2025 · Ahmed, Tennie & Magri 2025 · Li et al. 2025 · Tandon et al. 2025 · Hou et al. 2025 · Čindrak et al. 2026 · Antoncich et al. 2026 · Kobayashi & Motome 2026 · Huang et al. 2021 · Huang, Kueng & Preskill 2020 (classical shadows) · Dambre et al. 2012 · Corsi 2009 · Patton 2011 · Hansen et al. 2011 · Harvey et al. 1997 · Diebold & Mariano 1995 · Bollerslev 1986 · Jaeger 2001 · Heber et al. 2009.